

OPERATION ALPHA

MARITIME SECURITY & TRANSIT CORRIDOR PROTOCOL

WESTERN NAVAL COMMAND • HEADQUARTERS MUMBAI

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All units assigned to Operation Alpha shall ensure receipt acknowledgment within 4 hours of promulgation.

EXECUTIVE SUMMARY & MANDATE

STRATEGIC OBJECTIVES & STRATEGIC CONTEXT

2.1 Strategic Context

In accordance with Western Fleet Operational Order 042/26, Task Force 54 is deployed to maintain continuous surveillance, freedom of navigation, and maritime sovereignty across the Eastern Arabian Sea corridor. Heightened regional maritime activity requires reinforced presence along the 200 NM Exclusive Economic Zone (EEZ).

2.2 Primary Mission Objectives

1. Establish an impenetrable multi-layer surface and subsurface screening cordon around merchant transit channels.
2. Enforce strict Emission Control (EMCON Alpha) along sensitive tactical corridors to deny electronic signal exploitation.
3. Maintain real-time cryptographic audit trail of all distributed operational intelligence using post-quantum verifiable credentials.
4. Integrate shore-based coastal radar with airborne UAV continuous synthetic aperture radar (SAR) coverage.

STRATEGIC DIRECTIVE: Zero compromise on cryptographic chain-of-custody. Any leaked physical or electronic brief shall be traced immediately via forensic watermark lineage.

TASK FORCE 54 ORGANIZATION

FLEET COMPOSITION & COMMAND HIERARCHY

3.1 Task Force Composition

Unit Name	Callsign	Role / Class	Commanding Officer
INS Vikramaditya (R33)	VIKRAM-ALPHA	Carrier Strike Flagship	Capt. R. Deshmukh, VSM
INS Visakhapatnam (D66)	SENTINEL-BRAVO	Guided Missile Destroyer	Capt. V. K. Nair
INS Mormugao (D67)	PATROL-CHARLIE	Guided Missile Destroyer	Capt. A. Sengupta
INS Khanderi (S22)	HUNTER-ZULU	Attack Submarine (SSK)	Cdr. S. Ramanathan
UAV Squad Alpha-03	DRONE-ALPHA	Long-Range Surveillance	Lt. Cdr. M. Farooq
FIC-04 Squadron	INTERCEPT-04	Fast Interceptor Craft	Lt. P. K. Sharma

3.2 Command Channels

Operational command resides with Flag Officer Commanding Western Fleet (FOCWF), embarked onboard INS Vikramaditya. Secondary tactical command is delegated to INS Visakhapatnam in the event of EMCON Alpha signal isolation.

AREA OF OPERATIONS & SECTORS

GEOGRAPHIC BOUNDARIES & PATROL BOXES

4.1 Tactical Sector Coordinates

Sector Designation	North-West Bound	South-East Bound	Assigned Primary Unit
Sector Alpha (Outer)	19°30'N, 71°45'E	18°15'N, 72°30'E	INS Vikramaditya (R33)
Sector Bravo (Approach)	19°05'N, 72°20'E	18°40'N, 72°50'E	INS Visakhapatnam (D66)
Sector Charlie (Transit)	18°35'N, 72°30'E	17°10'N, 73°10'E	INS Mormugao (D67)
Box Zulu (Subsurface)	17°50'N, 72°15'E	17°10'N, 72°50'E	INS Khanderi (S22)
Air Corridor Delta	19°45'N, 72°10'E	19°10'N, 72°55'E	UAV-Alpha-03 Surveillance

NAVIGATION ADVISORY: All naval vessels shall maintain minimum 5 NM standoff from designated commercial of fshore gas production platforms.

HYDROGRAPHIC & ACOUSTIC INTEL

BATHYMETRY & THERMAL LAYER PROFILE

5.1 Arabian Sea Bathymetric Contours

The continental shelf extends approximately 80 NM westward from Mumbai coastline before dropping steeply past the 200m isobath to abyssal depths exceeding 2,500m. This gradient creates distinct acoustic propagation zones.

5.2 Sound Velocity Profile (SVP)

- Surface Duct Layer: 0 - 45 meters (Temp 28.5°C, sound speed 1542 m/s).
- Thermocline Zone: 45 - 180 meters (Rapid sound velocity decrease to 1498 m/s).
- Deep Sound Channel (SOFAR Axis): Below 800 meters (Optimal long-range acoustic detection).

ACOUSTIC ADVANTAGE: Subsurface unit INS Khanderi is advised to patrol below the 60m thermocline boundary to evade surface active sonar detection.

VESSEL READINESS MATRIX

MATERIEL READINESS, FUEL & WEAPONS STATE

6.1 Fleet Readiness Status (B-24 Hour Audit)

Unit	Displacement	Propulsion State	Fuel (F-76)	Weapons Readiness
INS Vikramaditya	45,400 tonnes	100% Operational	94% Full	MiG-29K Combat Ready (16 airfram
INS Visakhapatnam	7,400 tonnes	100% Operational	88% Full	BrahMos + Barak-8 (Full loadout)
INS Mormugao	7,400 tonnes	100% Operational	91% Full	BrahMos + Barak-8 (Full loadout)
INS Khanderi	1,775 tonnes	100% Operational	98% Battery	Varunastra Heavy Torpedoes ready
FIC-04 Squadron	45 tonnes	Twin Waterjets 100%	95% Full	12.7mm HMG + Spike Missiles

All propulsion and combat management suites are certified by Naval Dockyard Mumbai quality assurance inspection teams.

AVIATION & UAV FLIGHT PLAN

RECONNAISSANCE CORRIDORS & AEW PATROLS

7.1 Air Surveillance Schedule

Continuous airborne maritime domain awareness will be maintained through synchronised UAV and fixed-wing sorties from INS Hansa (Goa) and carrier air wing.

Air Asset	Mission Profile	Altitude	Loiter Endurance	Data Link
UAV-Alpha-03	Maritime Radar Sweep	FL 240 (24,000 ft)	14.5 Hours	PQC Encrypted DDL (C-Band)
Kamov Ka-31	Airborne Early Warning	FL 120 (12,000 ft)	3.2 Hours	Link-II Tactical Mesh
P-8I Neptune	Long Range ASW / C4I	FL 310 (31,000 ft)	8.0 Hours	SATCOM Ku-Band Encrypted

AIR DEFENSE INTEGRATION: Friendly air assets shall squawk Mode 5 Level 2 IFF cryptography at all times within the Western Fleet Air Defense Identification Zone.

SUBSURFACE WARFARE & ACOUSTICS

PASSIVE SURVEILLANCE & BUOY GRID

8.1 Anti-Submarine Warfare (ASW) Screen

Subsurface surveillance architecture employs a hybrid passive hydrophone array tethered along the continental shelf edge, supported by tactical sonobuoy fields deployed by maritime patrol aircraft.

8.2 Acoustic Sonobuoy Fields

- Pattern Alpha-Echo: 12x DIFAR passive buoys deployed along 72°30'E meridian.
- Pattern Bravo-Zulu: 8x VLAD deep-water acoustic buoys covering Mumbai approach canyon.
- Acoustic Intercept Log: Monitored 24/7 by Western Fleet Acoustic Intelligence Center (AIC).

RADIO SILENCE NOTICE: INS Khanderi shall not break surface or radiate electromagnetic signals unless receiving coded VLF burst priority 1 emergency dispatch.

TACTICAL COMMS ARCHITECTURE

POST-QUANTUM FREQUENCY ALLOCATIONS

9.1 Multi-Band Naval Communications Matrix

Band / System	Frequency Range	Primary Usage	Cryptographic Standard
EHF MilSat (GSAT-7)	44.0 - 46.0 GHz	Fleet Command Uplink	NIST ML-KEM-768 / FIPS 203
SHF Link-II Mesh	7.25 - 8.40 GHz	Inter-Ship Tactical Data	AES-256-GCM + PQC Kyber
VHF Marine Tactical	156.0 - 174.0 MHz	Close Formation Maneuver	Secured Frequency Hopping
VLF Subsurface	16.0 - 24.0 kHz	Submarine Emergency Order	One-Time Pad Acoustic

All primary and secondary data streams are signed with ML-DSA-65 post-quantum zero-trust credentials.

EMISSION CONTROL (EMCON) ROE

RADIO SILENCE MATRIX & SIGNATURE CONTROL

10.1 EMCON Operational States

State	Radar / Active RF	Radio / Voice	Satellite Uplink	Tactical Posture
EMCON Alpha	COMPLETELY SILENT	SILENT (Rx Only)	SILENT (Rx Only)	Full Stealth / Denied Environmen
EMCON Bravo	Directional LPI Only	Coded Burst Only	Scheduled Bursts	Restricted Tactical Transit
EMCON Charlie	Active Radar 3D Sweep	Encrypted Voice OK	Continuous Uplink	Standard Peacetime Escort

INCIDENT PROTOCOL: If any vessel experiences unintended RF radiation during EMCON Alpha, an automatic tel
emetry breach incident report is submitted to Fleet HQ ledger.

COMMERCIAL SHIPPING MANAGEMENT

TRANSIT CORRIDORS & BUFFER PROTOCOLS

11.1 De-confliction with Commercial Marine Traffic

Over 14,000 commercial cargo, container, and tanker vessels transit the Western Indian seaboard annually. Clear separation protocols ensure civilian maritime traffic remains undisturbed while tactical operations proceed.

11.2 Safety Buffer Parameters

- Commercial Traffic Separation Scheme (TSS): Minimum 3 NM lateral separation for task group vessels.
- Offshore Oil & Gas Platforms: Strict 500m UNCLOS safety zone enforced with 2 NM radar screening.
- JNPT / Mumbai Port Approach: Coordinated with Mumbai Vessel Traffic Management System (VTMS).

RULES OF ENGAGEMENT (ROE)

GRADUATED DEFENSIVE RESPONSE POSTURE

12.1 Graduated Escalation Ladder

Level	Threat Threshold	Authorized Defensive Measure	Command Approval Required
Tier 1	Unidentified vessel within 15 NM	Radar illumination, VHF challeng	Officer of the Watch (OOW)
Tier 2	Non-responsive within 8 NM	Acoustic hailing, Warning flares	Executive Officer / Captain
Tier 3	Aggressive maneuvering < 3 NM	Warning shots across bow	Flag Officer Commanding Fleet
Tier 4	Hostile intent / Imminent attack	Kinetic neutralization of weapon	National Command Authority

SOVEREIGN MANDATE: Hostile action against Indian flag vessels within territorial waters shall be met with proportionate self-defense under Article 51 of UN Charter.

LOGISTICS & FLEET REPLENISHMENT

RAS STATIONS, MAINTENANCE & FUEL SUPPLY

13.1 Replenishment At Sea (RAS) Schedule

Fleet tanker INS Deepak (A50) will execute liquid and solid replenishment at designated underway rendezvous points every 72 hours.

RAS Station	Rendezvous Coordinates	Primary Customer	Supplies Transferred
Point Romeo-1	18°45'N, 72°15'E	INS Vikramaditya (R33)	Aviation Fuel (Avcat), F-76, Rat
Point Romeo-2	18°10'N, 72°35'E	INS Visakhapatnam & Mormugao	Diesel F-76, Fresh Water, Spares
Emergency Port 1	INS Kadamba (Karwar Base)	All Surface Units	Deep-water berths, Munitions dep
Emergency Port 2	Mormugao Enclave (Goa)	Fast Interceptor Craft	Rapid turnaround, engine refit

CYBER DEFENSE & PQC LEDGER

ZERO-TRUST FORENSIC PROVENANCE ENGINE

14.1 Zero-Trust Architecture & Cryptographic Provenance

NAV-TRAC X provides cryptographic defense against physical leak extraction and unauthorized brief duplication. Each distributed briefing package is watermarked steganographically at the server layer prior to recipient rendering.

14.2 Cryptographic Key Specifications

- Post-Quantum Key Encapsulation: NIST ML-KEM-768 (Kyber) for payload transit encryption.
- Post-Quantum Digital Signature: NIST ML-DSA-65 (Dilithium) for immutable tamper audit.
- Hash Chain Topology: Merkle-tree structured immutable audit blocks replicated across 6 command nodes.

AUDIT COMPLIANCE: Every document open, decrypt, print, and export event is committed as an immutable block on the naval ledger.

ANNEX A: TACTICAL SIGNAL CODES

RAPID FLASH MESSAGE & BREVITY CODES

A.1 Fleet Brevity Code Glossary

Code Word	Operational Meaning	Action Required by Units
FLASH ALPHA	EMCON radio silence enforced imm	Shut down non-essential transmit
RADAR DARK	Surveillance radar switch to sil	Activate optronic infrared senso
BLUE SHIELD	All crypto keys rotated to next	Confirm PQC certificate handshak
VIPER STRIKE	Unidentified fast surface contac	Deploy UAV thermal tracking
SANCTUARY	Proceed to emergency diversion b	Acknowledge via VLF beacon

ANNEX B: CONTINGENCY DIVERSION

PROJECT SEABIRD KARWAR & GOA ENCLAVE

B.1 Project Seabird (Karwar) Facilities

INS Kadamba Karwar serves as the strategic deep-water refuge base for capital ships with covered submarine dry-docks, underground munitions bunkers, and independent tactical communications grid.

B.2 Emergency Entry Coordinates

- Fairway Channel Buoy: 14°48'30"N, 74°06'15"E (Channel depth 13.5m CD).
- Harbor Control VTS: Calling Channel 12 VHF (Encrypted Tactical).
- Medical Facility: INHS Patanjali 150-bed tertiary trauma care.

ANNEX C: DISTRIBUTION & AUDIT

AUTHORIZED RECIPIENT ROSTER & SECURITY CLEARANCE

C.1 Authorized Recipient Register

Recipient Rank & Name	Unit / Command	Clearance Level	Access Mode
Cdr. Arjun Mehta	Directorate of Naval Operations	Level 4 (Top Secret)	Full Authorization
Lt. Cdr. Sunita Rao	Western Naval Command (HQ)	Level 4 (Top Secret)	Full Authorization
Lt. Priya Singh	Eastern Fleet Intelligence	Level 3 (Secret)	View Only (Secure)
Capt. V. K. Nair	INS Visakhapatnam (D66)	Level 4 (Top Secret)	Full Authorization
Capt. A. Sengupta	INS Mormugao (D67)	Level 4 (Top Secret)	Full Authorization
Cdr. S. Ramanathan	INS Khanderi (S22)	Level 4 (Top Secret)	EMCON Air-Gapped

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